

Prefix Capacities and Separation Cuts for Shared-Kernel Termination-Time Hiding

Companion toolbox to *Scheduling and Compiler Techniques for Metadata-Hiding Overlay Lookups*

Short note (January 2026)

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Abstract

This note is a compact addendum to the termination-time (stop-time) channel in *Scheduling and Compiler Techniques for Metadata-Hiding Overlay Lookups* [1]. The main paper optimizes over padding kernels (Kernel-TV-Pad) [1]. Here we instead characterize useful *projections* of that kernel polytope—deadlines and average stop-time laws—via closed forms and reusable cuts.

We provide three mechanically reusable facts. (1) **Prefix capacities:** for every deadline t , a sharp closed-form characterization of the maximal achievable probability of stopping by t under a TV budget, computable by a greedy fractional-knapsack routine. (2) **Not sufficient:** tight per-deadline bounds do not characterize global feasibility; we give a minimal rational counterexample at horizon $T = 3$. (3) **Membership & separation:** an LP $\text{TV}_{\min}(Q)$ certifies whether a proposed average stop-time law Q is implementable at a given budget and returns a violated linear inequality when it is not. We omit large artifacts and give reconstruction-oriented specifications.

Optional library. Appendices record additional structure mined from earlier drafts: a transport interpretation for the worst-case objective and two explicit $T = 3$ “library facts.”

1 Context, use-cases, and model

When to use this note (instead of just running Kernel-TV-Pad). Kernel-TV-Pad [1] is the right tool when you want a numeric optimum for a concrete instance. The results here are useful when you want *fast, explainable structure* in the lower-dimensional space of stop-time laws: (i) quick deadline bounds and sanity checks, (ii) a cheap tuning surface for choosing deadlines/budgets before running a full LP, (iii) a cutting-plane workflow over averages Q via membership/separation cuts (Section 4), and (iv) a lemma-pack that an automated reader can reuse without re-deriving the full kernel LP.

Discrete success times and shared delay-only kernels. Fix a discrete horizon $\{1, \dots, T\}$. A success time $S \in \{1, \dots, T\}$ is drawn from one of two laws μ_0, μ_1 (“worlds” / “secrets”). A padding rule is a *shared delay-only kernel* K : an upper-triangular row-stochastic matrix with entries

$K(s, t) = \mathbb{P}[Y = t \mid S = s]$ for $t \geq s$ and $K(s, t) = 0$ for $t < s$. The induced stop-time laws are

$$P_b(t) = \sum_{s \leq t} \mu_b(s) K(s, t), \quad b \in \{0, 1\}.$$

We also use the average stop-time law

$$Q := \frac{1}{2}(P_0 + P_1).$$

Privacy budget (TV). If the adversary observes only Y , the optimal distinguishing advantage is

$$\text{TV}(P_0, P_1) = \frac{1}{2} \sum_{t=1}^T |P_0(t) - P_1(t)|.$$

We enforce approximate hiding by $\text{TV}(P_0, P_1) \leq \Delta$.

Mass and charge. Define the average *mass* and signed *charge* on success times:

$$m(s) := \frac{1}{2}(\mu_0(s) + \mu_1(s)), \quad w(s) := \mu_0(s) - \mu_1(s),$$

so $\sum_s m(s) = 1$ and $\sum_s w(s) = 0$.

2 Prefix capacities: sharp bounds for deadlines

Fix a deadline $t < T$ and write the average prefix CDF $F_Q(t) := \mathbb{P}[Y \leq t]_{\text{avg}} = \sum_{u \leq t} Q(u)$.

2.1 The prefix-capacity LP

Only rows $s \leq t$ can contribute to the event $\{Y \leq t\}$ under a delay-only kernel. For $t < T$, define the *prefix capacity*

$$\text{cap}_t(\Delta) := \max \left\{ \sum_{s \leq t} m(s) y_s : 0 \leq y_s \leq 1, \left| \sum_{s \leq t} w(s) y_s \right| \leq \Delta \right\}. \quad (1)$$

2.2 Sharp deadline theorem

Theorem 2.1 (Prefix capacity is tight). *For every $t < T$,*

$$\max_{\substack{K \text{ shared delay-only} \\ \text{TV}(P_0, P_1) \leq \Delta}} F_Q(t) = \text{cap}_t(\Delta).$$

Proof. Fix a shared delay-only kernel K and define, for each $s \leq t$, $y_s := \mathbb{P}[Y \leq t \mid S = s] = \sum_{u \leq t} K(s, u)$. Then $0 \leq y_s \leq 1$ and

$$F_Q(t) = \sum_{s \leq t} m(s) y_s, \quad F_0(t) - F_1(t) = \sum_{s \leq t} w(s) y_s.$$

Since $\{1, \dots, t\}$ is an event, $|F_0(t) - F_1(t)| \leq \text{TV}(P_0, P_1) \leq \Delta$, so (y_s) is feasible for (1) and $F_Q(t) \leq \text{cap}_t(\Delta)$.

Conversely, given any feasible $(y_s)_{s \leq t}$, define a kernel that outputs only at times t and T : for $s \leq t$, set $K(s, t) = y_s$ and $K(s, T) = 1 - y_s$; for $s > t$, set $K(s, T) = 1$. Then $F_Q(t) = \sum_{s \leq t} m(s) y_s$ and

$$\text{TV}(P_0, P_1) = |F_0(t) - F_1(t)| = \left| \sum_{s \leq t} w(s) y_s \right| \leq \Delta,$$

so the bound is achievable. □

2.3 Greedy computation (fractional knapsack)

If $|\sum_{s \leq t} w(s)| \leq \Delta$, then $\text{cap}_t(\Delta) = \sum_{s \leq t} m(s)$ (take $y_s = 1$). Otherwise, to reduce the prefix charge into $[-\Delta, \Delta]$, decrease some y_s from 1. Decreasing y_s by α removes mass $\alpha m(s)$ and removes charge $\alpha w(s)$, so the mass cost per unit removed charge is $m(s)/|w(s)|$. Thus the optimum is obtained by removing charge starting from indices with largest $|w(s)|/m(s)$ and stopping fractionally on the last index.

Pseudocode (one deadline). Handle $w(s) = 0$ by treating $|w(s)|/m(s) = 0$.

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Input: arrays m[1..t], w[1..t], budget Delta
Initialize y[s] = 1 for all s <= t
c = sum_{s <= t} w[s]
if |c| <= Delta: return cap = sum_{s <= t} m[s], y

need = |c| - Delta
Sort indices s <= t by decreasing r[s] = |w[s]| / m[s]
for s in sorted order:
    if w[s] == 0: continue
    removable = |w[s]|
    take = min(need, removable)
    y[s] = 1 - take / removable
    need -= take
    if need == 0: break

cap = sum_{s <= t} m[s] * y[s]
return cap, y

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3 Tight prefix bounds are not sufficient

Theorem 2.1 yields necessary inequalities for any achievable average law Q :

$$F_Q(t) \leq \text{cap}_t(\Delta) \quad \forall t < T. \quad (2)$$

A tempting relaxation saturates each prefix and takes differences:

$$F_{\text{env}}(t) := \text{cap}_t(\Delta), \quad Q_{\text{env}}(t) := F_{\text{env}}(t) - F_{\text{env}}(t-1), \quad F_{\text{env}}(0) := 0.$$

This envelope respects every inequality (2) but can be globally infeasible because a single shared kernel must satisfy *all* deadlines at once.

Proposition 3.1 (Minimal envelope gap at $T = 3$). *Let $T = 3$, $\Delta = \frac{1}{20}$, and*

$$\mu_0 = \left(\frac{19}{60}, \frac{2}{5}, \frac{17}{60} \right), \quad \mu_1 = \left(\frac{4}{15}, \frac{1}{2}, \frac{7}{30} \right).$$

Then $\text{cap}_1(\Delta) = \frac{7}{24}$ and $\text{cap}_2(\Delta) = \frac{89}{120}$, so

$$Q_{\text{env}} = \left(\frac{7}{24}, \frac{9}{20}, \frac{31}{120} \right).$$

Yet Q_{env} is infeasible at budget Δ : its minimum required TV satisfies $\text{TV}_{\min}(Q_{\text{env}}) = \frac{1}{10} > \Delta$. Equivalently, Q_{env} violates the valid inequality

$$\frac{124}{315}Q(1) + \frac{2}{9}Q(2) \leq \frac{89}{540}, \quad \text{i.e.} \quad 1488Q(1) + 840Q(2) \leq 623.$$

(The inequality is a dual certificate returned by $\text{TV}_{\min}$ in Section 4.)

4 Membership and separation via $\text{TV}_{\min}(Q)$

Let $\mathcal{Q}_{\Delta} \subseteq \mathbb{R}^T$ be the set of all achievable average stop-time laws Q under shared delay-only kernels with $\text{TV}(P_0, P_1) \leq \Delta$. Prefix inequalities are necessary but not sufficient (Proposition 3.1). This section gives a compact LP that decides membership and returns separating cuts.

4.1 A membership LP

For a candidate average law Q° , define

$$\text{TV}_{\min}(Q^\circ) := \min\{\text{TV}(P_0, P_1) : \exists \text{ shared delay-only } K \text{ with } \frac{1}{2}(\mu_0 K + \mu_1 K) = Q^\circ\}.$$

This is a linear program.

LP specification (mechanically implementable). Introduce variables $K(s, t) \geq 0$ ($1 \leq s \leq t \leq T$), $P_0(t), P_1(t) \geq 0$, and $u(t) \geq 0$ for $t = 1, \dots, T$, with constraints:

$$\begin{aligned} \text{(row sums)} \quad & \sum_{t=s}^T K(s, t) = 1 \quad \forall s, \\ \text{(pushforward)} \quad & P_b(t) = \sum_{s \leq t} \mu_b(s) K(s, t) \quad \forall t, b \in \{0, 1\}, \\ \text{(match target)} \quad & \frac{1}{2}(P_0(t) + P_1(t)) = Q^\circ(t) \quad \forall t, \\ \text{(linearize TV)} \quad & u(t) \geq P_0(t) - P_1(t), \quad u(t) \geq -(P_0(t) - P_1(t)) \quad \forall t. \end{aligned}$$

Minimize $\frac{1}{2} \sum_{t=1}^T u(t)$. The optimum equals $\text{TV}_{\min}(Q^\circ)$.

Theorem 4.1 (Membership and separation). *Fix Δ .*

1. **Membership:** $Q^\circ \in \mathcal{Q}_{\Delta}$ if and only if $\text{TV}_{\min}(Q^\circ) \leq \Delta$.
2. **Separation:** If $Q^\circ \notin \mathcal{Q}_{\Delta}$, LP duality returns an explicit vector $g \in \mathbb{R}^T$ such that $g \cdot Q \leq \sup\{g \cdot Q' : Q' \in \mathcal{Q}_{\Delta}\}$ holds for all $Q \in \mathcal{Q}_{\Delta}$ but is violated by Q° .

Remark 4.2 (Cut extraction in one paragraph). *In practice, one solves the primal LP above and reads the dual multipliers on the “match target” constraints $\frac{1}{2}(P_0(t) + P_1(t)) = Q^\circ(t)$. Those multipliers define the separating direction g in Theorem 4.1. If $\text{TV}_{\min}(Q^\circ) > \Delta$, the dual optimal solution is a certificate: it yields a concrete linear inequality in Q that is valid for $\mathcal{Q}_{\Delta}$ and violated by Q° (as in Proposition 3.1).*

5 Reconstruction specification (short)

We omit large artifacts; the results are intended to be reconstructible from the specification:

1. Compute m, w from (μ_0, μ_1) .
2. For each deadline t , compute $\text{cap}_t(\Delta)$ by the greedy fractional-knapsack routine.
3. Decide feasibility of any candidate Q° at budget Δ by solving $\text{TV}_{\min}(Q^\circ)$ and, if infeasible, extracting a dual cut.

6 Discussion

Prefix capacities give sharp per-deadline bounds for shared-kernel termination-time hiding. The envelope counterexample shows global feasibility is genuinely polyhedral: sharp local bounds do not characterize the projected feasible set. Finally, $\text{TV}_{\min}$ provides a practical bridge between full kernel optimization (Kernel-TV-Pad [1]) and coarse relaxations: it certifies designs and produces explicit separating cuts.

A Optional: transport equivalence for the worst-case objective

This appendix records additional structure for the worst-case overhead objective $\mathbb{E}_{\max}[Y] = \max\{\mathbb{E}[Y \mid 0], \mathbb{E}[Y \mid 1]\}$. It is not needed for the main story (Sections 2–4).

Shared-kernel minimax optimization (KI). For a budget $\tau \in [0, \tau_{\max}]$ where $\tau_{\max} := \text{TV}(\mu_0, \mu_1)$, define

$$\text{OPT}_{KI}(\tau) := \min_{K \text{ shared delay-only}} \max_{b \in \{0,1\}} \mathbb{E}[Y \mid b] \quad \text{s.t.} \quad \text{TV}(\mu_0 K, \mu_1 K) \leq \tau.$$

Let $\tau_{\min} := \min_K \text{TV}(\mu_0 K, \mu_1 K)$ be the minimum achievable leakage.

Unbalanced transport (UT- τ). Let $d := \mu_0 - \mu_1$ and define $P := \{i : d_i > 0\}$, $N := \{j : d_j < 0\}$ with $p_i := d_i$ for $i \in P$ and $q_j := -d_j$ for $j \in N$. A UT plan has variables $f_{ij} \geq 0$ ($i \in P, j \in N$) and unmatched masses $u_i \geq 0$ ($i \in P$), $v_j \geq 0$ ($j \in N$) such that

$$\sum_{j \in N} f_{ij} + u_i = p_i, \quad \sum_{i \in P} f_{ij} + v_j = q_j,$$

and unmatched mass $r := \sum_{i \in P} u_i = \sum_{j \in N} v_j$ satisfies $r \leq \tau$. Let $m(i, j) := \max\{i, j\}$ and scaling factors $A_i^{(b)} := \mu_b(i)/p_i$ ($i \in P$) and $B_j^{(b)} := \mu_b(j)/q_j$ ($j \in N$). Define

$$E_b(f, u, v) := \sum_{i \in P, j \in N} (A_i^{(b)} + B_j^{(b)}) m(i, j) f_{ij} + \sum_{i \in P} A_i^{(b)} i u_i + \sum_{j \in N} B_j^{(b)} j v_j + \sum_{s: d_s=0} \mu_b(s) s.$$

Set $\text{OPT}_{UT}(\tau) := \min_{(f, u, v): r \leq \tau} \max\{E_0(f, u, v), E_1(f, u, v)\}$.

Theorem A.1 (KI $\Leftrightarrow$ UT (exact leakage)). *For all $\tau \in [0, \tau_{\max}]$ (and taking $\text{OPT}_{KI}(\tau) = \text{OPT}_{UT}(\tau) = +\infty$ for $\tau < \tau_{\min}$),*

$$\text{OPT}_{KI}(\tau) = \text{OPT}_{UT}(\tau).$$

Proposition A.2 (Value function properties). *$\text{OPT}_{KI}(\tau)$ is finite on $[\tau_{\min}, \tau_{\max}]$, equals $+\infty$ on $[0, \tau_{\min})$, and is nonincreasing, convex, and piecewise-linear on $[\tau_{\min}, \tau_{\max}]$.*

B Optional: two $T = 3$ library facts

B.1 Perfect hiding at $T = 3$: closed-form feasible averages

Let $T = 3$ and write $\mu_0 = (a_1, a_2, a_3)$, $\mu_1 = (b_1, b_2, b_3)$. Let $\mathcal{Q}_{KI}$ denote the set of average stop-time laws $Q = \frac{1}{2}(\mu_0 K + \mu_1 K)$ achievable with perfect hiding ($\mu_0 K = \mu_1 K$).

Theorem B.1 (Closed form for $\mathcal{Q}_{KI}$ when $T = 3$). *Assume $\mu_0 \neq \mu_1$.*

1. (**Triangle case:** $a_1 = b_1$.) *Then $\mathcal{Q}_{KI}$ is exactly*

$$Q(1) \geq 0, \quad Q(2) \geq 0, \quad Q(1) + Q(2) \leq a_1, \quad Q(3) = 1 - Q(1) - Q(2).$$

2. (**Edge case:** $a_1 \neq b_1$.) *Then every $Q \in \mathcal{Q}_{KI}$ satisfies $Q(1) = 0$, so $\mathcal{Q}_{KI} = \{(0, q, 1 - q) : q \in [0, q_{\max}]\}$, where: if $a_2 = b_2$, then $q_{\max} = a_2$; otherwise let $r := -(a_1 - b_1)/(a_2 - b_2)$. If $r \leq 0$, then $q_{\max} = 0$; if $r > 0$, then $q_{\max} = \min\{a_1 + a_2 r, a_2 + a_1/r\}$.*

B.2 Intermediate leakage: an explicit strict chord gap

Proposition B.2 (Strict chord gap at $T = 3$). *Let $T = 3$ and $\mu_0 = (4, 3, 1)/8$, $\mu_1 = (1, 2, 5)/8$. Then $\text{TV}(\mu_0, \mu_1) = 1/2$ and perfect hiding is feasible with an optimal perfect-hiding target $Q_*^{KI} = \delta_3$. At budget $\tau = 1/4$, the endpoint-mixture baseline yields worst-case mean $z_{\text{mix}} = 11/4$. However, the shared kernel*

$$K(1, 1) = 1/3, \quad K(1, 3) = 2/3; \quad K(2, 2) = 1; \quad K(3, 3) = 1$$

induces $\text{TV}(P_0, P_1) = 1/4$ and achieves $\max\{\mathbb{E}[P_0], \mathbb{E}[P_1]\} = \mathbb{E}[P_1] = 8/3 < 11/4$.

References

- [1] *Scheduling and Compiler Techniques for Metadata-Hiding Overlay Lookups*. Wiki: [[2026.01.22 - Mathematics: Scheduling and Compiler Techniques for Metadata-Hiding Overlay Lookups]]. Draft v0.67 (2026-01-21). Anonymous submission; author list omitted.